

Abstract

We construct a parameter-free geometric prediction for the leptonic CP-violating phase from the holonomy of the compact hyperbolic 3-manifold $M_{\text{PMNS}} = m003(-2, 3)$ (`OrientableClosedCensus[1]`, volume 0.9814, $H_1 = \mathbb{Z}/5$). The twist angles $\varphi(\gamma) = \text{Im} \log \lambda(\gamma)$ (principal branch) of two loxodromic holonomy words yield:

$$\delta_{\text{HFG}} = (\pi + \varphi(\text{aaB}) + \varphi(\text{baa})) \bmod 360^\circ = 195.91^\circ,$$

numerically consistent with the PDG 2024 best fit 197.0° (0.55% discrepancy, no continuously tuned parameters). The branch choice of the leading π is fixed geometrically by the orientation of the axis triple: $\det[\hat{n}(\text{aa}), \hat{n}(\text{aaB}), \hat{n}(\text{baa})] = +0.079 > 0$. The manifold's trace field $K_{\text{PMNS}} = \mathbb{Q}(\sqrt{-3})$ (imaginary quadratic) provides a qualitative geometric distinction between the large leptonic and suppressed quark-sector CP phases within the HFG construction. The prediction is falsifiable by Hyper-Kamiokande and DUNE.

CP Violation from Twist Angles of Hyperbolic Holonomy: A Parameter-Free Prediction of the Leptonic CP Phase

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1 Introduction

The leptonic CP-violating phase δ_{PMNS} is the least well-measured parameter of the Standard Model flavor sector. The PDG 2024 best fit is $\delta = 197.0^\circ$ with a 1σ uncertainty of approximately $\pm 23^\circ$; Hyper-Kamiokande and DUNE are expected to reduce this to $\pm 5^\circ$ precision. No first-principles derivation of δ exists.

In this paper we present evidence that δ is encoded in the twist angles of the holonomy representation of a single compact hyperbolic 3-manifold, $M_{\text{PMNS}} = m003(-2, 3)$, with no continuously tuned parameters. The result is part of the Hyperbolic Flavor Geometry (HFG) framework [6], in which the PMNS mixing matrix and charged lepton masses are also obtained from the geometry of M_{PMNS} .

All computations use SnapPy [2] version 3.3.2 at 150-bit precision.

2 The Manifold and Its Holonomy

The manifold $M_{\text{PMNS}} = m003(-2, 3)$ has volume 0.9814, $H_1 = \mathbb{Z}/5$, and Chern-Simons invariant $1/4$. Its invariant trace field is $K_{\text{PMNS}} = \mathbb{Q}(\sqrt{-3})$ (imaginary quadratic) [3], with ring of integers $\mathbb{Z}[\omega]$, $\omega = e^{2\pi i/3}$. The CKM companion manifold $m006(-5, 2)$ has trace field $\mathbb{Q}(\sqrt{17})$ (real quadratic); since distinct invariant trace fields imply non-commensurability for arithmetic hyperbolic 3-manifolds [4], the two manifolds lie in distinct commensurability classes.

The holonomy representation $\rho : \pi_1(M_{\text{PMNS}}) \rightarrow \text{PSL}(2, \mathbb{C})$ is computed via SnapPy's `polished_holonomy()` at 150-bit precision. For a loxodromic element $\gamma \in \pi_1(M_{\text{PMNS}})$, the eigenvalue of maximal modulus $\lambda(\gamma)$ satisfies $|\lambda(\gamma)| > 1$, and the *twist angle* is $\varphi(\gamma) = \text{Im} \log \lambda(\gamma) \in (-\pi, \pi]$, where the principal branch of the logarithm is used throughout.

The axis direction $\hat{n}(\gamma) \in S^2$ is extracted from the Pauli decomposition of $\log \rho(\gamma)$:

$$\log \rho(\gamma) = \frac{\eta}{2}I + \frac{i}{2}(n_x\sigma_x + n_y\sigma_y + n_z\sigma_z), \quad \hat{n} = \frac{(n_x, n_y, n_z)}{\|(n_x, n_y, n_z)\|},$$

where $\eta = \ell(\gamma) + i\varphi(\gamma)$ is the complex translation length.

3 Word Triple Selection and Methodology

The word triple $\{\text{aa}, \text{aaB}, \text{baa}\}$ was selected from an exhaustive scan of all primitive words in the generators $\{a, b\}$ of $\pi_1(M_{\text{PMNS}})$ up to word length 5 (484 primitive words before conjugacy reduction, yielding 34 distinct geodesic classes). The selection criterion was minimisation of the

Frobenius fitness $\mathcal{F} = |||Q| - |U_{\text{PMNS}}|_{\text{PDG}}||_F$ under the Borel (lower-triangular) overlap construction (the N -factor of the Iwasawa decomposition), with the smearing parameter σ optimised over $[0.1, 2.0]$ by Nelder–Mead for each triple. The triple $\{\text{aa}, \text{aaB}, \text{baa}\}$ achieves fitness $\mathcal{F} = 0.005087$, the minimum within the scanned ensemble ($p < 10^{-4}$ against 50,000 Haar-random unitaries).

The CP phase formula in Theorem 1 uses the same word triple but accesses only the A -factor (twist angles) of the Iwasawa decomposition—a strictly separate quantity from the N -factor used for the mixing matrix. The formula was not selected to match δ_{PDG} ; it was identified from the word triple already fixed by the mixing matrix construction. This independence is what removes any free parameter from the CP phase prediction.

4 The CP Phase Formula

Theorem 1 (Geometric CP-phase computation). *Let ρ be the polished holonomy of $M_{\text{PMNS}} = m003(-2, 3)$. Using the word triple $\{\text{aa}, \text{aaB}, \text{baa}\}$ fixed by the Borel mixing construction (Section 3), the twist-angle combination*

$$\delta_{\text{HFG}} = (\pi + \varphi(\text{aaB}) + \varphi(\text{baa})) \bmod 360^\circ = 195.91^\circ$$

is numerically consistent with the PDG 2024 best fit $\delta_{\text{PDG}} = 197.0^\circ$ (discrepancy 1.09° , 0.55%). No continuously tuned parameters enter once the word triple is fixed.

Computational verification. The twist angles from SnapPy `polished_holonomy()` at 150-bit precision, using the principal branch of $\log$:

$$\varphi(\text{aaB}) = -176.73^\circ, \quad \varphi(\text{baa}) = -167.36^\circ.$$

Then:

$$180^\circ + (-176.73^\circ) + (-167.36^\circ) = -164.09^\circ \equiv 195.91^\circ \pmod{360^\circ}.$$

The branch choice of the leading π is fixed geometrically:

$$\det[\hat{n}(\text{aa}), \hat{n}(\text{aaB}), \hat{n}(\text{baa})] = +0.079 > 0$$

(right-handed triple, selecting $+\pi$ rather than $-\pi$). Reproducible via `conda run -n sage python reproduce/cp_reproduce.py` at [7]. $\square$

Remark 2. The formula is invariant under conjugation of each word in $\pi_1(M_{\text{PMNS}})$: conjugate elements share the same eigenvalues and hence the same twist angles. The result is also insensitive to the choice of generator set, as SnapPy’s polished holonomy is computed from the geometric triangulation, not a presentation.

Remark 3. The word triple $\{\text{aa}, \text{aaB}, \text{baa}\}$ achieves the minimum Frobenius fitness identified in the scanned ensemble under the Borel PMNS construction [5]. The CP phase formula uses only the A -factor (twist angles) of the Iwasawa decomposition $\text{PSL}(2, \mathbb{C}) = KAN$, complementing the N -factor (Borel) used for the mixing matrix.

5 Geometric Distinction of CP Sectors

Proposition 4 (Trace field dichotomy). *The PMNS and CKM manifolds have distinct invariant trace fields: $K_{\text{PMNS}} = \mathbb{Q}(\sqrt{-3})$ (imaginary quadratic, discriminant -3) and $K_{\text{CKM}} = \mathbb{Q}(\sqrt{17})$ (real quadratic, discriminant $+17$). For arithmetic hyperbolic 3-manifolds, the sign of the discriminant determines whether holonomy elements are generically twisted loxodromic (disc < 0) or purely hyperbolic (disc > 0) [4].*

Verification. For M_{PMNS} : established in [3]. For $M_{\text{CKM}} = m006(-5, 2)$: $\text{tr}(\rho(aa)) = 3 - \sqrt{17}$, satisfying $x^2 - 6x - 8 = 0$ (discriminant $68 = 4 \cdot 17$), verified at 150-bit precision. $\square$

Within the HFG construction, this arithmetic distinction provides a qualitative geometric account of the large leptonic CP phase ($\delta = 195.91^\circ$, twisted loxodromic holonomy) versus the suppressed quark CP violation (purely hyperbolic holonomy, geometric $J = 0$ at zeroth order). The observed non-zero quark Jarlskog invariant ($J_{\text{exp}} \approx 3 \times 10^{-5}$) requires corrections beyond the zeroth-order construction.

6 Comparison with PDG and Experimental Status

The PDG 2024 fit gives $\delta_{\text{PDG}} = 197^\circ$ with 1σ range approximately $[174^\circ, 220^\circ]$. The geometric prediction $\delta_{\text{HFG}} = 195.91^\circ$ lies within the 1σ range and differs from the central value by 1.09° (0.55%).

Table 1: Geometric prediction vs observed leptonic CP phase.

Quantity	HFG prediction	PDG 2024
δ_{PMNS}	195.91°	197.0°
$\varphi(\text{aaB})$	-176.73°	—
$\varphi(\text{baa})$	-167.36°	—
Discrepancy	1.09° (0.55%)	—
Continuous fit parameters	0	—

7 Falsifiability

The prediction $\delta_{\text{HFG}} = 195.91^\circ$ is falsifiable:

- (1) **Hyper-Kamiokande** will measure δ to $\sim 5^\circ$ precision. A value outside $[180^\circ, 215^\circ]$ at 3σ would falsify Theorem 1.
- (2) **DUNE** provides an independent measurement with comparable precision.
- (3) The formula uses words aaB and baa from the triple fixed by the Borel mixing construction. If a different triple were found to give better PMNS fitness, the CP phase formula would need to be re-evaluated.

Open problems. The quark CP phase δ_{CKM} from the HFG construction gives 72° (PDG: 68° , 5.9%), consistent at geometric precision. The mechanism producing the non-zero J_{exp} from the geometric zeroth-order $J = 0$ remains the most pressing open problem in the framework.

8 Conclusions

We have computed the leptonic CP-violating phase from the twist angles of the holonomy of $M_{\text{PMNS}} = m003(-2, 3)$: $\delta_{\text{HFG}} = 195.91^\circ$, numerically consistent with PDG 2024 (0.55% discrepancy, no continuously tuned parameters). The branch choice is fixed geometrically by the orientation of the axis triple. The result is one component of the Hyperbolic Flavor Geometry framework [6]; the same manifold encodes the PMNS mixing matrix and charged lepton mass ratios. The imaginary/real trace field dichotomy (Proposition 4) provides a qualitative geometric account of the CP asymmetry between sectors. The prediction is falsifiable by Hyper-Kamiokande and DUNE.

Data Availability

All scripts and data at [7]. Canonical reproduction: `conda run -n sage python reproduce/cp_reproduce.py`

Acknowledgments

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References

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